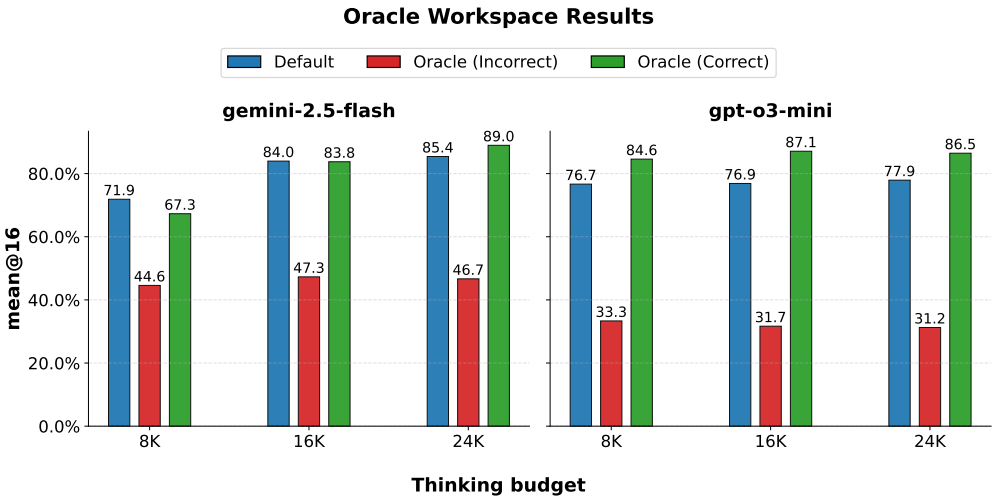

Figure 8 AIME 2025: Anchoring bias due to +ve and −ve examples: With **PDR** we compare three selection policies for the summary: Random-k, Oracle-Incorrect (all k candidates are incorrect), and Oracle-Correct (all k candidates are correct), evaluated on both `gemini-2.5-flash` and `o3-mini`. Across all thinking budgets, admitting only incorrect candidates into the summary yields a pronounced drop in accuracy, whereas admitting only correct candidates improves over the Random-k baseline. The degradation under Oracle-Incorrect is markedly larger for `o3-mini` than for `gemini-2.5-flash`, indicating weaker self-verification in `o3-mini`.

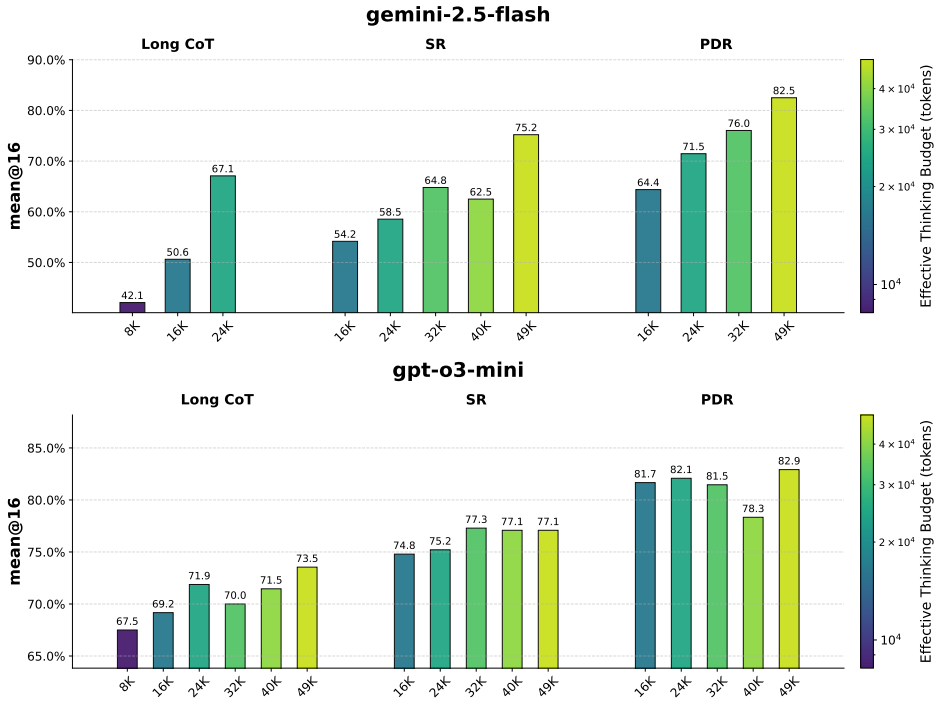

Figure 9 AIME 2025: Iterative improvement beats single-pass long-CoT at matched sequential budgets. The x-axis reports B_{seq}: the thinking tokens consumed along the accepted path of the iterative chain, plus any distilled summary that conditions the next step. Tokens spent on unused parallel proposals are excluded, so B_{seq} serves as a latency proxy. At comparable B_{seq}, both **SR** and **PDR** outperform the single-pass long CoT baseline, with **PDR** yielding the largest gains by converting additional total compute (via parallelism) into accuracy without increasing per-call context.

Additionally, we show two **SR** variant results in Table 1, where error analysis followed by solution generation leads to improvements on `o3-mini` without any affect on the sequential token budget B_{seq}.

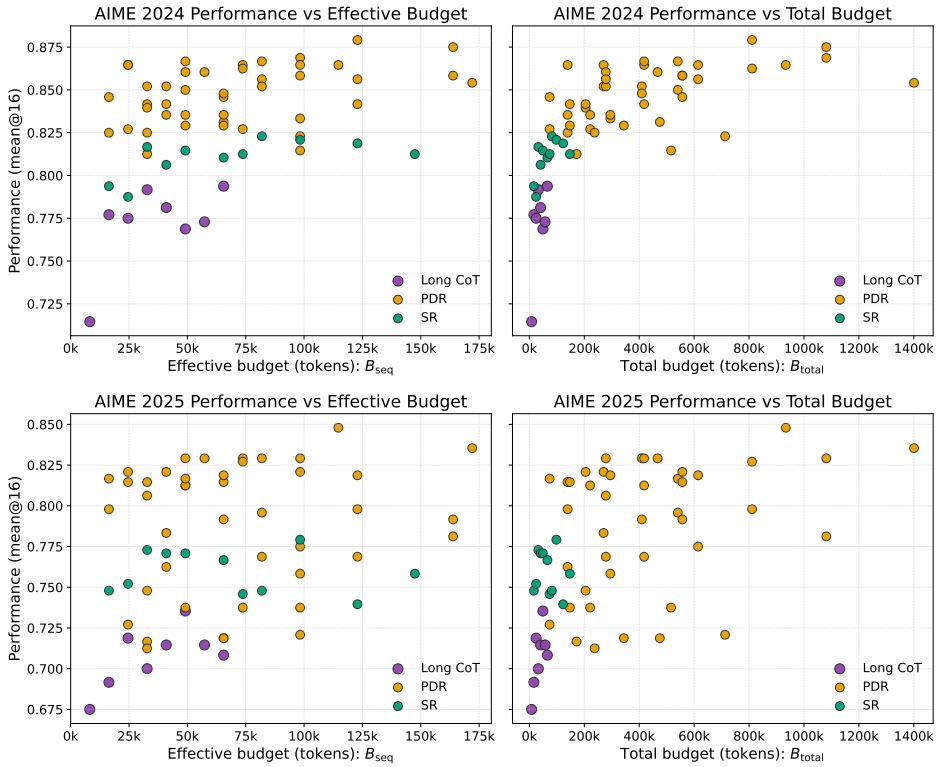

Figure 10 Token Budgets comparison: We plot all the different configurations for Long CoT, **SR** and **PDR** operators for both B_{seq} and B_{total} token budgets for `o3-mini`. For both B_{seq} and B_{total}, **PDR** forms the pareto-frontier and gives consistent gains over Long CoT and **SR**.

E Mechanics of Improvement operator: Source of accuracy gain

Under the default **PDR** setting, `gemini-2.5-flash` misses 4 AIME 2024 questions. We probe how additional parallel compute affects the performance on these four AIME questions. We compare a 4-round schedule (less compute) to a 5-round, wider schedule (more compute). Accuracy (fraction correct over 16 seeds) changes as follows: Q1: $0.4375 \to 0.625$ (gain), Q2: $0.0625 \to 0$ (drop), Q3: $0.1875 \to 0.1875$ (no change), Q4: $0 \to 0$ (no change). In the high-compute setting (first round width M_1=32), number of correct drafts among 32 is: Q1: 3/32, Q2: 0/32, Q3: 3/32, Q4: 0/32. This breakdown clarifies how **PDR** can (or cannot) improve with additional rounds:

Table 4 PDR hard-case analysis for `gemini-2.5-flash`. We compare a 4-round schedule (*less compute*: 16 generations in round 1) to a 5-round, wider schedule (*more compute*: 32 generations in round 1). Accuracies are fraction correct over 16 seeds. "Round-1 hits" counts how many of the 32 first-round drafts are already correct ((*more compute setting*)

	Round-1 hits	Accuracy (over 16 seeds)		Δ	Interpretation
Question	(correct/32)	Less compute	More compute	(More − Less)	
Q1	3/32	7/16 (0.4375)	10/16 (0.6250)	+3/16 (+0.1875)	Gain
Q2	0/32	1/16 (0.0625)	0/16 (0.0000)	−1/16 (−0.0625)	Drop
Q3	3/32	3/16 (0.1875)	3/16 (0.1875)	0	Flat
Q4	0/32	0/16 (0.0000)	0/16 (0.0000)	0	Flat

1. When a round-1 draft is correct, **PDR** improves if two things happen: (i) the correct evidence is carried into the summary $C^{(1)}$ (i.e., high recall in the distillation operator $\mathcal{D}$); and (ii) the refine step uses that evidence to update the answer. If $\mathcal{D}$ drops or down-weights the signal amid conflicting drafts, later rounds cannot exploit it. The Q1 gain from 4→5 rounds suggests both steps succeeded; the flat Q3 curve despite

3/32 correct drafts points to a verification/refinement gap.

2. **Verification among many distractors (Q1/Q3).** Even when correct drafts are present, round-1 mixes a small number of correct drafts with many incorrect ones (here, 3 vs. 29). The summary must surface signals that distinguish correctness. This is where **PDR**'s distill step should act as a verifier-aware aggregator.

3. **Recovery when no draft is correct (Q2/Q4).** If round 1 has 0/32 correct, the summary still needs to extract useful structure (partial progress, contradictions, eliminated avenues) that increases the chance of success in round 2+. The refine step should then expand diversity informed by these cues. The mild regression on Q2 with more compute is consistent with summary drift: the distillation over-weights a wrong pattern, and subsequent rounds reinforce it. The no-change on Q4 may suggest either a capability ceiling or the case that even more generations are required in round 1.

Overall, the analysis here points to some gaps in the core skills required to carry out **PDR** - verification, refinement, and diversity. Improving the model along each of these skills will not only improve **PDR** but also any situation where a multiplicative combination of these skills is required.